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High-Volume Prescribers Do Not Drive Medicare Prescriptions of Schedule II Opioids

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To the Editor

Researchers have suggested that the opioid overdose epidemic¹ is primarily driven by small groups of prolific prescribers and “corrupt pill mills.”^{2,3} For example, the California Workers’ Compensation Institute found that 1% of prescribers accounted for one-third of schedule II opioid prescriptions and 10% accounted for 80% of prescriptions.⁴ This propagates a message that opioid overprescribing is a problem of a small group of high-volume prescribers, while general use is likely safe and effective. Medicare data provide the opportunity to address whether such prescribing patterns occur across a national population.

Methods

We examined individual prescriber data from the 2013 Medicare Part D (prescription drug coverage) claims dataset created by the Centers for Medicare and Medicaid Services.⁵ Part

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Author Contributions: Dr. Chen had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Chen, Humphreys, Lembke

Acquisition of data: Chen, Shah

Analysis and interpretation of data: Chen

Drafting of the manuscript: Chen

Critical revision of the manuscript for important intellectual content: All authors

Statistical analyses: Chen

Administrative, technical, or material support: Shah

Study supervision: Humphreys, Lembke

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Promotional Image: Figure 2

Tweet: Opioid Prescribing Distribution in Medicare: Not just a few bad apples.

D covers ~68% of the ~50 million people on Medicare, the federal insurance program for Americans who have certain disabilities or are 65 or older.

For each prescriber National Provider Identifier (NPI) number (N=808,020), the data identify each drug prescribed, total number of claims, and total costs. Each NPI includes location and specialty of practice. The data represent 1,188,393,892 claims for \$80,941,763,731. We focused on schedule II opioid prescriptions containing hydrocodone, oxycodone, fentanyl, morphine, methadone, hydromorphone, oxymorphone, meperidine, codeine, opium, or levorphanol.

We calculated the cumulative claims for schedule II opioids from the top individual prescribers (sorted by number of claims) relative to the total claims for all prescribers. For comparisons, we repeated this for prescription *costs*, for all drugs, and for each state.

Results

Figure 1 reports which provider specialties account for the most opioid drug claims. Figure 2 reports the concentration of drug claims amongst the most prolific individual prescribers. Respective California Workers' Compensation data⁴ are included. Notably, the top 10% of Medicare prescribers account for a smaller proportion of opioid claims (56.7%) than for all Medicare prescriptions and for the California Workers' Compensation prescribers. Minimal regional variation is observed across provider states, with per state values ranging from 56.6% to 57.7%. Excluding hydrocodone (schedule III prior to 2014) yields similar trends with the same top three prescribing specialties and 57.9% of claims from the top 10% of prescribers.

Comment

The data studied represent a comprehensive national population of Medicare Part D prescribers, but do not necessarily reflect providers' complete practices, patient factors (e.g., comorbidities and prescription indications), or medication dosing to inform morphine equivalents. With those cautions, two important findings are evident.

Opioid prescriptions are concentrated in specialty services in Pain, Anesthesia, and Physical Medicine and Rehabilitation (PM&R). By sheer volume however, total prescriptions are dominated by general practitioners (Family Practice, Internal Medicine, Nurse Practitioners, and Physician Assistants).

Contrary to the California Worker's Compensation data showing a small subset of prescribers accounting for a disproportionately large percentage of opioid prescribing, Medicare opioid prescribing is distributed across many prescribers and is, if anything, less skewed than all drug prescribing. The trends hold up across state lines, with negligible geographic variability. Figure 2 does show greater skewing for total drug *costs* of Medicare opioid claims, with 78% accounted for by 10% of prescribers. This could be selection of more expensive formulations or higher doses prescribed.

The distribution of any social phenomena has some degree of skewing similar to an “80/20 rule” (e.g., 20% of the population controls 80% of the wealth).⁶ As of 2013 however, these data argue that opioid prescribing is no more skewed than other prescriptions, reflecting a widespread practice relatively indifferent to individual doctors, specialty or region. High-volume prescribers are not responsible for the high national volume of opioid prescriptions. Efforts to curtail national opioid overprescribing must address a broad swath of prescribers to be effective.

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References

1. Jones CM, Mack KA, Paulozzi LJ. Pharmaceutical Overdose Deaths, United States, 2010. *JAMA*. 2013; 309:657–659. [PubMed: 23423407]
2. Rosenau AM. Guidelines for opioid prescription: The devil is in the details. *Ann Intern Med*. 2013; 158:843–844. [PubMed: 23567867]
3. Betses M, Brennan T. Abusive prescribing of controlled substances—a pharmacy view. *N Engl J Med*. 2013; 369:989–91. [PubMed: 23964897]
4. Ireland J, Johnson G. Prescribing Patterns of Schedule II Opioids in California Workers’ Compensation. 2011
5. Medicare Provider Utilization and Payment Data: Part D Prescriber. 2015. at <<http://www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/Medicare-Provider-Charge-Data/Part-D-Prescriber.html>>
6. Reed WJ. The Pareto, Zipf and other power laws. *Econ Lett*. 2001; 74:15–19.

Schedule II Opioid Claims By Provider Specialty

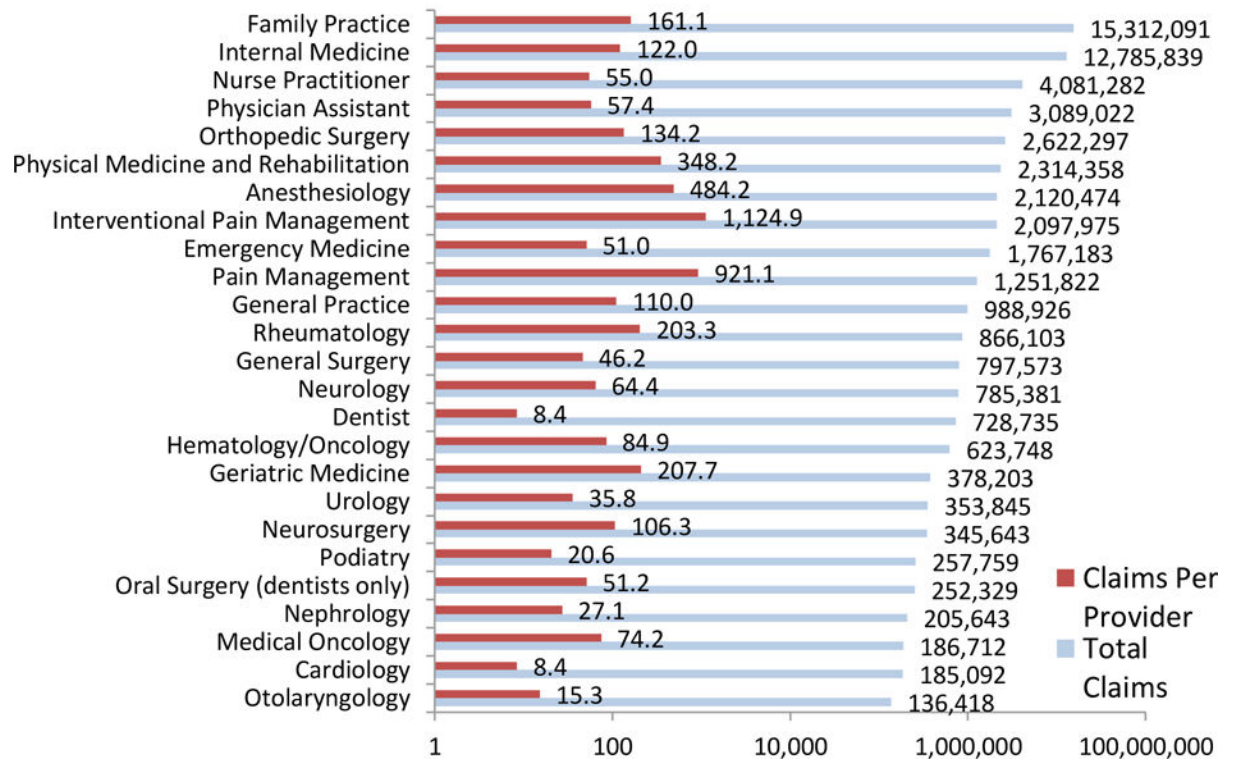


Figure 1.

Top 25 provider specialties by total Medicare Part D claims for schedule II opioids in 2013.

Values reported on logarithmic scale.

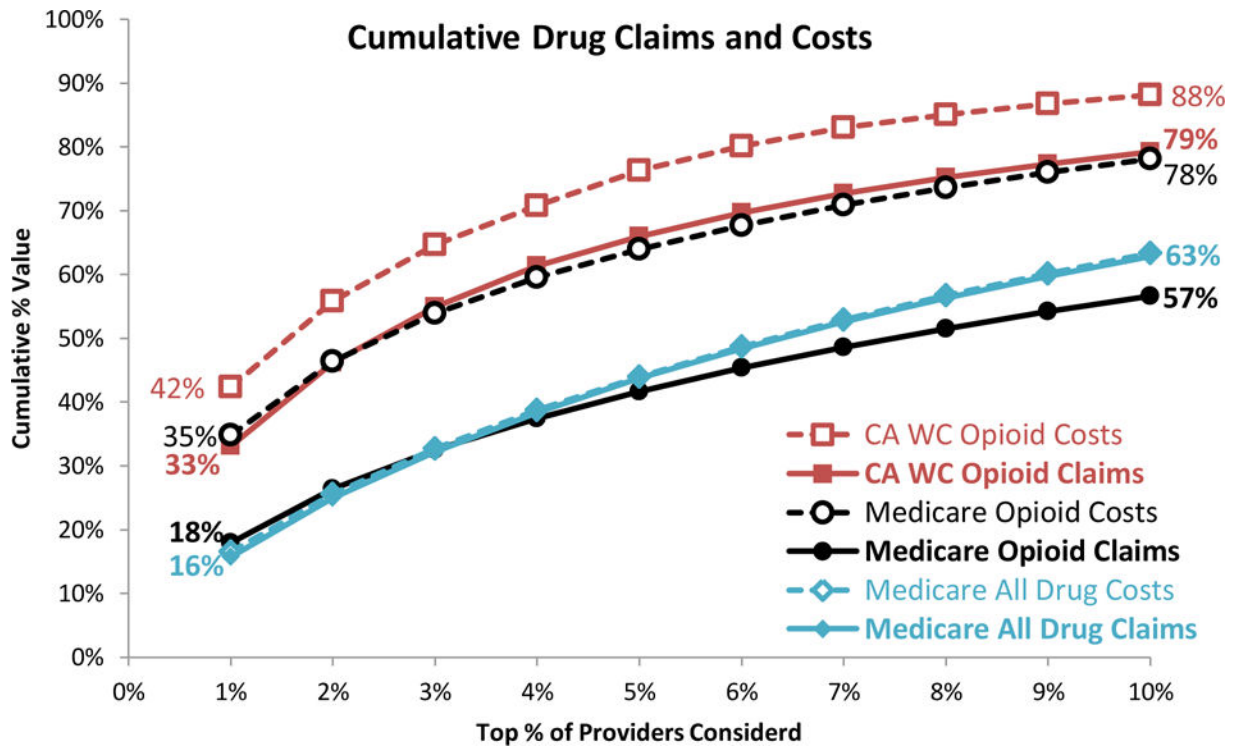


Figure 2.

Cumulative percent claims and costs for the top ten percent of prescribers for different populations. For example, 1% of California workers' compensation (CA WC) providers incur 42% of their schedule II opioid costs. Note: The Medicare All Drug Claims curve overlaps and obscures the respective Costs curve.